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AI Reflection Coach



TEACHING & RESEARCH DESIGN SCENARIO 1 - SMART INSULIN PEN WITH ML COMPONENT THE STORY SmartDose GmbH builds a connected insulin pen for diabetic patients. The pen logs every injection and syncs data to a mobile app. 9 developers, running Scrum with 2-week sprints - working well for 3 years.

The CEO returns from a digital health conference with a new plan: "We add an ML model that analyses each patient's blood glucose history and automatically recommends the next insulin dose. This makes the pen truly smart."

What changes immediately:

- The ML model is trained on data, not coded. Its output is probabilistic - it will sometimes be wrong, and a wrong insulin dose can be life-threatening.
- Training requires blood glucose histories, meal logs, and physical activity data from real patients. This is highly sensitive personal health data.
- EU law (MDR) classifies this AI as a medical device. The model must be clinically validated and approved before any real patient acts on its output.
- Every patient is different. A dose that is correct for one person may be dangerous for another - the model must learn per-patient patterns and

must be monitored continuously for safety.

- Nobody on the current team has ML or data science experience.

The CEO wants a prototype in 6 months and regulatory approval in 18 months.

TASK 1 - ANALYSE a. Which Scrum elements still work? Which create new problems? (Bullet points are fine) Scrum Element Still works? Why / Why not?
2 – week sprints Definition of Done Backlog prioritization Sprint Review
Demo Velocity tracking

TASK 2 - DECIDE Three situations. For each: pick one option and justify your choice. "We would do both" is not an answer.

SITUATION A - WHO DO YOU HIRE FIRST? You have budget for exactly 2 new hires. You need all four of these roles: (1) ML Engineer - builds and trains the model (2) Clinical Validator - tests whether the model is medically safe (3) Data Privacy Officer - ensures GDPR compliance for patient data (4) Regulatory Specialist - navigates MDR approval with the authorities

a. Which 2 do you hire first? b. What risk do you consciously accept by waiting on the other 2? c. What is the earliest point in the project where the missing roles become a critical bottleneck?

SITUATION B - WHICH DATA DO YOU USE? To train the model you need patient data. Two hospitals offer data: (1) Hospital A: 50,000 patient records, large and diverse dataset. Problem: the data was collected before AI training was a use case - patients never explicitly consented to this. (2) Hospital B: 5,000 patient records, full informed consent for AI use. Problem: 5,000 records may

not be enough to train a reliable model. A model trained on too little data can fail dangerously on edge cases

a. Which dataset do you use - A, B, or neither? b. What are the consequences of your choice for patient safety and c. for the project timeline?

SITUATION C - WHAT DO YOU TELL THE CEO? The CEO insists: "Prototype shown to real patients in 6 months." EU regulatory approval (MDR) takes a minimum of 18 months. You cannot legally show the model's dosage output to real patients before approval - doing so would be a criminal offence.

You have three options: (1) Push back on the CEO: delay the patient-facing demo to month 18. Risk: loss of investor confidence, possible funding cut. (2) Build a "shadow mode" prototype: the model runs in the background and generates recommendations, but patients never see them. A doctor reviews the hidden output for safety research only. (3) Launch in a non-EU country first where regulations are less strict, gather real-world data, use it to speed up the EU approval.

a. Which option do you choose? b. What does your actual project timeline look like as a result?

TASK 3 - REFLECT In 2-3 sentences: a. Which of the three situations was hardest to decide and why? b. What would change your answer if one key assumption were different?



It looks like you're working on a complex scenario involving a connected insulin pen and machine learning, which presents numerous challenges. Let's delve into your analysis one part at a time.

TASK 1 - ANALYZE

For the Scrum elements, consider the following:

2-week sprints:

- **Still works?** Yes.
- **Why / Why not?** The sprints can still provide a structured way to iterate and deliver components, but the integration of ML increases unpredictability in the scope and outcomes.

Definition of Done:

- **Still works?** Partially.
- **Why / Why not?** While a Definition of Done is essential, it may need to evolve to include considerations for model validation and safety checks, which are crucial for this scenario.

Backlog prioritization:

- **Still works?** Yes.
- **Why / Why not?** Prioritization remains vital, but the criteria may shift significantly to account for regulatory requirements and model safety.

Sprint Review Demo:

- **Still works?** Partially.
- **Why / Why not?** While reviews maintain value, ensuring patient safety with a model in development adds layers of complexity to what can be demonstrated.

Velocity tracking:

- **Still works?** Partially.
- **Why / Why not?** The team may struggle to estimate velocity accurately due to the introduction of new roles and uncertainties related to ML development.

TASK 2 - DECIDE

SITUATION A: a. Which 2 do you hire first?

b. What risk do you consciously accept by waiting on the other 2?

c. What is the earliest point in the project where the missing roles become a critical bottleneck?

Reflect on what you think:

What specific considerations guided you toward your choices for hiring? Are there implications for team dynamics or project milestones with your decision?

TASK 3 - REFLECT

a. Which of the three situations was hardest to decide and why?

b. What would change your answer if one key assumption were different?

Consider these prompts:

Why did a specific situation challenge you more than others? What assumption could lead to a fundamentally different approach or outcome?

Feel free to share your thoughts on these tasks as you analyze this scenario!



For task 2. give me better solutions



I appreciate your openness to explore better solutions! Let's take a closer look at each situation in Task 2 and refine your approaches.

SITUATION A: WHO DO YOU HIRE FIRST?

a. Which 2 do you hire first?

Consider hiring both the **ML Engineer** and the **Regulatory Specialist** first. The ML Engineer is essential for developing the model, and the Regulatory Specialist is crucial for navigating the legal complexities involved.

b. What risk do you consciously accept by waiting on the other 2?

Accept that there could be significant delays in ensuring data privacy compliance and clinical validation processes if you delay hiring the Data Privacy Officer and Clinical Validator.

c. What is the earliest point in the project where the missing roles become a critical bottleneck?

The missing Clinical Validator becomes a bottleneck as soon as the model is ready for training and testing, ideally around the 5-6 month mark when validation needs arise.

SITUATION B: WHICH DATA DO YOU USE?

a. Which dataset do you use - A, B, or neither?

Choose **Hospital B**, despite its smaller size, because it has informed consent for AI use. This maintains ethical standards and legal compliance regarding patient data.

b. What are the consequences of your choice for patient safety?

By selecting Hospital B, you ensure that patient safety and legal compliance are prioritized, but the model's reliability may be compromised due to the limited data.

c. What are the consequences for the project timeline?

Using a smaller dataset may delay the model's readiness for clinical validation, extending the timeline for a functional prototype and pushing back regulatory approval processes.

SITUATION C: WHAT DO YOU TELL THE CEO?

a. Which option do you choose?

Choose **Option 2: Build a "shadow mode" prototype**. This allows you to gather valuable insights and data without violating regulations while prioritizing patient safety.

b. What does your actual project timeline look like as a result?

By building the shadow mode, you can still demonstrate some progress to the CEO within 6 months. However, you would need to account for additional time to integrate feedback and prepare for compliance to allow the model to be used safely in a patient-facing capacity.

What specific considerations led you to different answers, and have any constraints changed from your initial thoughts?



kann you give me an better answer for task 3



Let's refine your approach to Task 3 and generate more thorough and thoughtful responses based on the three situations.

TASK 3 - REFLECT

a. Which of the three situations was hardest to decide and why?

The hardest situation to decide was **SITUATION C** because it involves balancing ethical responsibilities, regulatory constraints, and the CEO's expectations. The pressure to deliver a prototype within six months collides with legal realities, making it challenging to maintain transparency while also satisfying stakeholders' requirements.

b. What would change your answer if one key assumption were different?

If the assumption that regulatory approval strictly prohibits any patient interaction with unapproved outputs were relaxed (for example, if provisional use in a controlled clinical setting could be allowed), I might reconsider choosing the shadow mode prototype. This could open up the possibility to gather real-world feedback more rapidly and adjust the model based on patient interactions, thus potentially accelerating the validation process.

How do these reflections resonate with your thinking? Are there any areas where you feel you could dive deeper into the implications of your choices?

Message the AI Coach...

